

Kryštof Przeczek

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PROFILE

I am a Computer Science student focused on graphics programming, real-time rendering, and simulation. I have experience building Vulkan renderers, path tracers, and physics simulations from scratch in C/C++. I have an interest in performance-critical systems combining computer science, mathematics, physics, and visual art.

SKILLS

- **Languages:** C, C++, C#, GLSL, HLSL, Slang (rasterisation and compute tasks), Java, and Python.
- **Graphics APIs:** OpenGL and Vulkan.
- **Tools:** Git, Unix terminal, RenderDoc, Xcode, and Radeon Developer Suite.
- **Engines & Frameworks:** C# FNA/MonoGame library, Unity, Godot, and GameMaker Studio 2.

PROJECTS

Magpie | *C, SDL, Vulkan, ImGui* <https://github.com/kryzp/magpie>
Vulkan renderer with hot-reloadable code and assets, async asset streaming, a fiber-based job system, physically-based materials, with a render graph and scene system for efficient GPU synchronization and geometry management.

Stick it Together | *FastAPI, Kotlin, React, ElevenLabs* <https://github.com/kryzp/sotonhack26backend>
Winner of SotonHack '26. 1v1 party game linking concepts via semantic similarity scoring. Built with Kotlin, React, and FastAPI; uses Gemini for scoring and ElevenLabs for real-time voice commentary.

Path Tracer | *C++* <https://github.com/kryzp/pathtracer>
Path tracer capable of physically-based metallic, lambertian, and refractive materials.

2D Finite-Element Solver | *Java* <https://github.com/kryzp/simple-finite-element-solver>
Computes beam deflection via stiffness matrices built from a 2D structure and boundary conditions.

2D Navier-Stokes Fluid Solver | *C++* <https://github.com/kryzp/simple-navier-fluid-sim>
Fluid simulation using iterative projection methods to solve for diffusion, advection and external forces.

EDUCATION

University of Southampton 2024–Present
Computer Science MEng

Hills Road Sixth Form College, Cambridge 2022–2024
Further Maths A, Maths A, Computer Science A, Physics B

WORK EXPERIENCE

Microsoft June 2022
Summer Work Placement

- Shadowed a senior software engineer to learn about the tools and algorithms used daily to ensure rapid response times in their high-speed communications network.

ACHIEVEMENTS

- **Programming Competitions:** Competed in Advent of Code (2022–Present), placing 1st in the school (out of 150) in Year 13, and 2nd in Year 12.
- **Olympiads:** Earned Silver in both the British Informatics Olympiad (BIO) and British Physics Olympiad (BPhO).
- **Project Euler:** Completed 104 problems independently, ranking me in the top 5% of all competitors.
- **Duke of Edinburgh's Award (DofE):** Refined skills in game development, swimming, and volunteering at a local library.

HOBBIES

- Analysing games by intercepting CPU-to-GPU data to study their graphics techniques.
- Experimenting with esoteric shaders on Shadertoy to create visually striking scenes.
- Sketching landscapes and painting wildlife digitally.
- Playing guitar, I'm a fan of post-rock and post-apocalyptic styles.
- Undertaking more hands-on projects like building a crossbow and carpentry.